

Experiment No.2

OBJECT: To show Geometric transformations- rotation, scaling and translation in image.

THEORY:

We can show the image using different types of geometric transformations based on the application to perform. Some of the geometric transformations are:-

1. Rotation: Image can be rotated in the same direction.

`imrotate(I, angle)` rotates image I by angle degrees in a counterclockwise direction around its center point. To rotate the image clockwise, specify a negative value for angle. `imrotate` makes the output image J large enough to contain the entire rotated image. `imrotate` uses nearest neighbor interpolation, setting the values of pixels in J that are outside the rotated image to 0 (zero).

2. Scaling: It scales the image accordingly.

`Imresize (A, scale)` returns image B that is scale times the size of A. The input image A can be a grayscale, RGB, or binary image. If A has more than two dimensions, `imresize` only resizes the first two dimensions. If scale is in the range [0, 1], B is smaller than A. If scale is greater than 1, B is larger than A. By default, `imresize` uses bicubic interpolation.

`Imresize (A,[numrows numcols])` returns image B that has the number of rows and columns specified by the two-element vector [numrows numcols].

3. Translation: It translates the image in the particular direction desired.

`imtranslate(A,translation)` translates image A by the translation vector specified in translation. If A has more than two dimensions and translation is a two-element vector, `imtranslate` applies a 2-D translation to A, one plane at a time.

PROGRAM:

```
clc;
```

```
clear all;
```

```
close all;

a=imread ('g.jpeg');

b=imrotate (a,45);

c=imrotate (a,180);

d=imrotate(a,-45);

subplot(2,2,1);imshow(a);title('color image');

subplot(2,2,2);imshow(b);title('image rotate by 45 degree (left rotation)');%+ will show left
rotation & non multiple of 90 will show black portion except image

subplot(2,2,3);imshow(c);title('image rotate by 180 degree');%it is in multiple of 90,so no black
portion is shown

subplot(2,2,4);imshow(d);title('image rotate by -45 degree(right rotation)');%-ve will show right
rotation
```

RESULT:

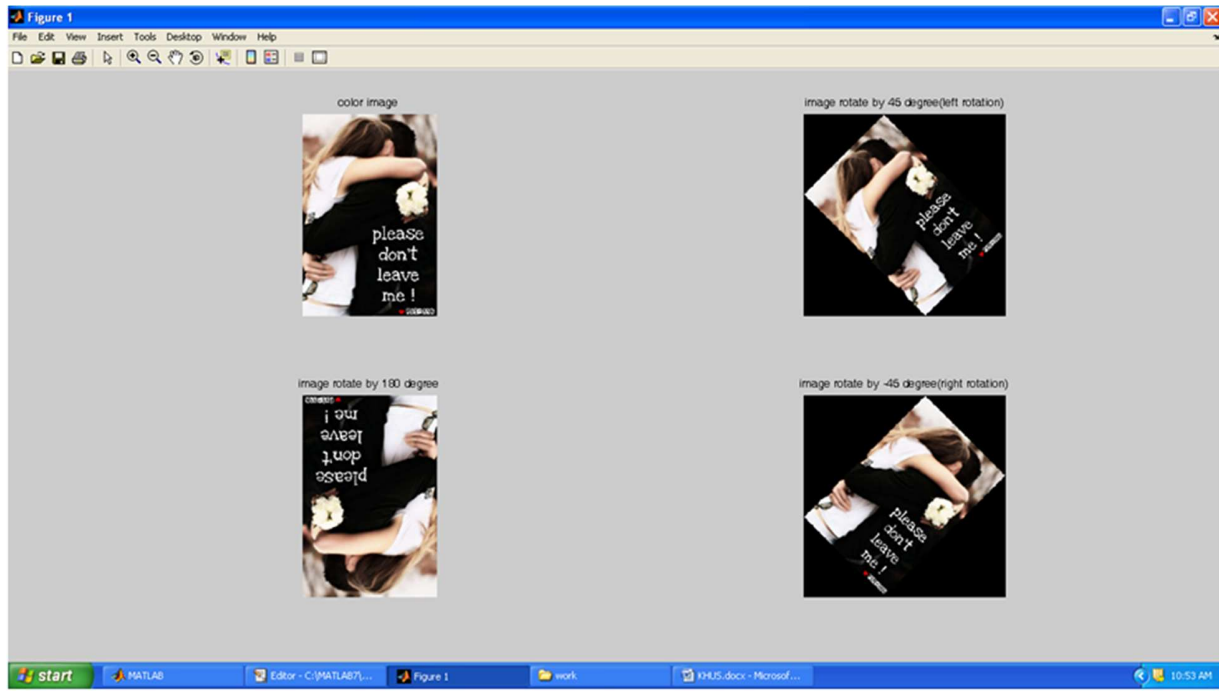


Figure 2.1: Rotated Images

```
clc;
```

```
clear all;
```

```
close all;
```

```
a=imread ('baby.jpg');
```

```
b=imresize (a,[250,250]); %direct command used to resize an given image
```

```
subplot(1,2,1);imshow(a);title('CUTE BABY');
```

```
subplot (1,2,2);imshow(b);title('CUTE BABY');
```



Figure 2.2: Resize Images

```
I = imread('cameraman.tif');
```

Display the image. The size of the image is 256-by-256 pixels. By default, `imshow` displays the image with the upper right corner at (0,0).

```
figure  
imshow(I)  
title('Original Image')
```

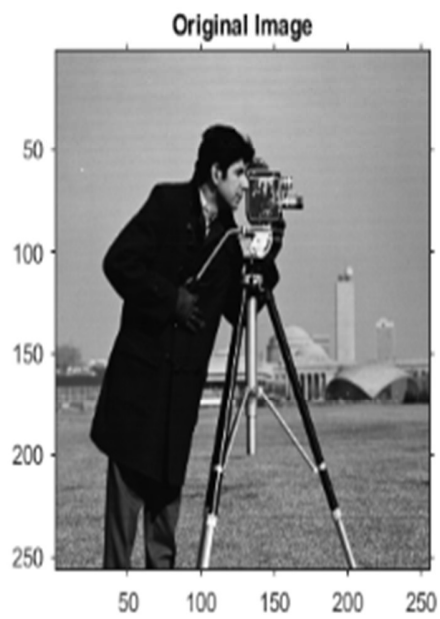


Figure 2.3: Original Images

```
J = imtranslate(I,[15, 25]);
```

Display the translated image. The size of the image is 256-by-256 pixels.

```
figure
imshow(J)
title('Translated Image')
```

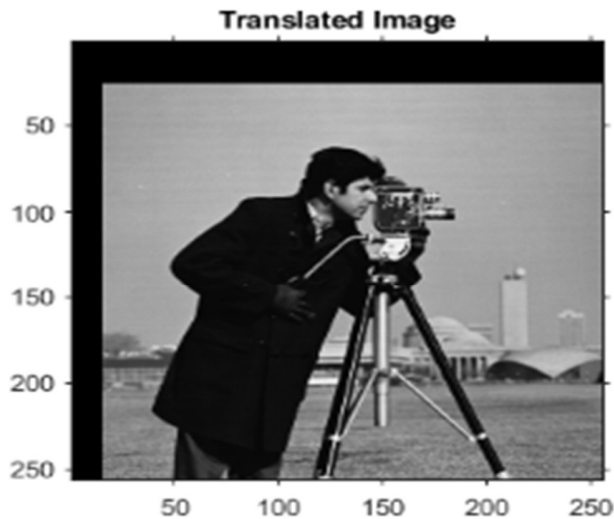


Figure 2.4: Translated Images

VIVA-VOCE QUESTIONS:

1. What do you mean by geometrical transformations of an image?
2. How can you show rotation in an image?
3. How can you show scaling or resizing in an image?
4. How can you show translation in an image without clipping its any part?
5. Can you rotate an image in anti-clockwise direction? If yes, how?

REFERENCES

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